

LedgerLens Product Blueprint and MVP PRD

Oct 4, 2026 · @Bhawna

How to read this

Recommendation in one line: position LedgerLens as the *explainable, reviewer-first* statement-underwriting layer for Indian lenders. Data extraction is becoming a commodity through Account Aggregators. LedgerLens should win on three things incumbents do poorly: trust in the statement, plain-language reasons for every number, and a review workflow that learns from credit officers.

Labels used throughout

Label	Meaning
[Confirmed]	Already built in the LedgerLens prototype, or a published fact with a source linked in this doc
[Assumption]	Our working choice where nothing is decided yet; to confirm with the business
[Reg/Legal]	Depends on RBI, Account Aggregator, data-protection or fair-lending rules; needs compliance counsel before build
[Approx.]	From general market knowledge, not verified against a source page in this doc

What the prototype already does [Confirmed]

- Reads digital PDF, scanned PDF, phone photo, password-protected PDF and CSV; several months and accounts in one upload.
- Runs 22 data-validation checks, including page order, running-balance arithmetic, duplicates and holder-name consistency.
- Sorts every transaction rules-first, then LLM, then a person in the Review queue, each with a confidence and method.
- Scores six weighted components into a 1000-point score with hard overrides, and gives a decision: APPROVE, APPROVE WITH CONDITIONS, REFER or DECLINE.
- Explains every score with ⓘ panels; keeps an activity log shared on the server; offers API v1 with SQLite storage, a background queue and versioned results.

Key assumptions to confirm first

1. **[Assumption]** First customers are **NBFCs and digital lenders** doing unsecured personal and small-business loans of ₹50,000 to ₹25 lakh. Banks come later, after security audits.
2. **[Assumption]** LedgerLens is sold **B2B to regulated lenders**. The lender owns the credit decision; LedgerLens recommends.
3. **[Assumption]** Data comes from **uploaded statements first, Account Aggregator second**. Most small-ticket and self-employed borrowers still share PDFs.
4. **[Reg/Legal]** Using an LLM on borrower data, storing statements, and making automated credit recommendations all need compliance review. See the regulatory register at the end.

1. Market research

Finding: extraction and categorisation are table stakes in India. Every serious vendor reads PDFs, scans, net banking and Account Aggregator (AA) data and claims high categorisation accuracy. Competition has moved to **fraud detection** (FinBox claims 15+ fraud types and 150+ rules) and **packaged scores**. Almost nobody competes on **explaining the decision to the credit officer** or on **learning from reviewer corrections**.

Method note. Vendor websites could not be opened from this research environment (blocked by the network proxy). Vendor claims below come from search-result summaries of the vendors' own pages and press, and are marked *claimed*. Verify each claim in a demo before relying on it.

Who is in the market

Player	Type	What they lead with	Notable claims	Source
Perfios (owns Karza since 2022)	Indian BSA + full underwriting suite	Scale and breadth: BSA, financial statements, KYC, fraud (Clari5), collections	1,000+ institutions; built-in checks for ePDF manipulation; 30M+ narrations a day categorised (<i>claimed</i>)	Perfios BSA, Inc42 on Karza

Player	Type	What they lead with	Notable claims	Source
FinBox BankConnect	Indian BSA for digital lenders	Fraud detection and speed	500+ parameters; 15+ fraud types, 150+ rule checks; document fingerprinting; 83% of applications in under 5 minutes (<i>claimed</i>)	FinBox BankConnect , FinBox fraud
Digitap	Indian BSA + AA Technology Service Provider	Multi-mode sourcing (net banking, PDF, AA) for fintech lenders	Certified AA TSP on Sahamati; KreditBee as a client (<i>claimed</i>)	Outlook Startup
ScoreMe	Indian BSA + packaged score, MSME focus	One report combining bank, GST, ITR and financial statements	Packaged score for the lender's pipeline (<i>claimed</i>)	CB Insights
Setu	AA gateway + insights API	One integration to many AAs, with income and expense insights on top	Insights API works on AA data and PDFs	Setu docs
Novel Patterns	Indian BSA inside a wider lending stack	Statement analysis as part of onboarding and decisioning	Details not verified	Unico Connect list
Plaid (US benchmark)	Open-banking network + cash-flow score	LendScore 1-99 from cash flow and network signals	Trained on ~1 billion transactions to predict 12-month default (<i>claimed</i>)	Plaid Consumer Report , LendScore blog

Player	Type	What they lead with	Notable claims	Source
Ocrolus (US benchmark)	Document AI + cash-flow analytics	Human-in-the-loop extraction; visual flags on edited documents	2,100+ document types (<i>claimed</i>)	Ocrolus
Inscribe (US benchmark)	Document-fraud specialist	Forensic + semantic + network checks; AI agents for fraud analysts	Bank statements were 59% of fraudulent documents it caught; AI-generated document fraud up 208% Apr-Dec 2025 (<i>claimed</i>)	Inscribe

Account Aggregator: the data rail everyone is moving to

- Sahamati figures reported for mid-2026: about **17 operational AAs**, **176 data providers (FIPs)**, **1,020+ data users (FIUs)**, **450M+ consents fulfilled** and **~295M linked accounts** (*search summary of Sahamati data*).
- **NBFCs drove ~60% of completed consents in FY25** ([Business Standard](#)).
- Implication: for borrowers whose bank is on AA, **raw data becomes free and clean**, and tamper risk falls sharply. Value shifts from *reading* statements to *interpreting* them. PDFs remain for banks not yet live on AA, for MSMEs and DSAs, and for borrowers who will not give AA consent.

Capability comparison

Legend: **Yes** = claimed by the vendor in sources above · **Likely** = typical for this vendor type, not verified [**Approx.**] · **—** = not seen in sources · LedgerLens column is [**Confirmed**] from the prototype.

Capability	Perfios	FinBox	Digitap	ScoreMe	Plaid (US)	Inscribe (US)	LedgerLens today
PDF + scanned upload	Yes	Yes	Yes	Yes	—	Yes	Yes (OCR in browser)
Net banking fetch	Yes	Yes	Yes	Likely	n/a	—	No

Capability	Perfios	FinBox	Digitap	ScoreMe	Plaid (US)	Inscribe (US)	LedgeLens today
Account Aggregator fetch	Likely	Yes	Yes (TSP)	Likely	n/a (own network)	—	No
Multi-account merge	Yes	Likely	Yes	Likely	Yes	—	Yes
Categorisation	Yes (30M/day)	Yes (500+ params)	Yes	Yes	Yes	—	Yes (rules → LLM → review)
Income vs P2P / self-transfer	Yes (salary)	Yes	Yes	Yes	Yes	—	Yes
EMI / obligation + FOIR	Yes (liability)	Yes	Yes	Yes	Likely	—	Yes
PDF tamper / document forensics	Yes	Yes (fingerprinting)	Likely	Likely	—	Yes (core)	Partial (balance arithmetic only)
Transaction fraud (circular, round-tripping)	Yes	Yes	Likely	Likely	—	Network checks	Yes (circular, structuring, pass-through)
Packaged score	Likely	Likely	Likely	Yes	Yes (LendScore)	Risk rating	Yes (1000 points + decision)
GST / ITR combined view	Yes	Likely	—	Yes	n/a	—	No
Plain-language explanation of each score	—	—	—	—	Partial (reason codes)	AI agent notes	Yes (① panels)
Human review queue that feeds back	—	—	—	—	—	Analyst workflow	Yes (review + counterparty rules)
Post-disbursal monitoring	Likely (via AA)	Likely	Likely	—	Yes	—	No

What this tells us

- 1. Do not compete on extraction breadth.** Perfios and FinBox have years of bank templates. LedgeLens should treat extraction as a utility: AA first, LLM fallback for unseen formats.
- 2. Fraud is the battleground in 2026.** Bank statements are the most-forged document, and AI-generated fakes are rising fast. Balance arithmetic alone is not enough; document forensics is a must for Phase 2.
- 3. The open gap is the credit officer's experience:** explanations, a review queue that learns, and a policy layer the lender can change without the vendor. None of the Indian leaders sell on this.

2. Product vision and positioning

Vision: every lending decision based on a bank statement is **fast, defensible and explainable**. A credit officer should understand in two minutes why a borrower is approved or refused, and should be able to defend that to a credit committee, an auditor or the borrower.

One-line positioning [Assumption]: *LedgerLens turns any bank statement into a decision a credit officer can trust and explain, and gets smarter every time an officer corrects it.*

The problem, by who feels it

Who	Problem today	Cost of the problem
Lender (business)	Statement analysis is a black box. Scores differ by vendor and nobody can say why. Policy changes need a vendor ticket.	Slow product launches; bad loans that "looked fine"; audit findings
Credit officer / underwriter	Gets a 40-tab Excel report. Re-checks salary, EMIs and bounces by hand. Cannot see which numbers are guesses.	20–45 minutes a file [Approx.]; inconsistent decisions between officers
Risk head	Cannot tell which rules or variables actually predict default; no feedback loop from repayment back to the score.	Policy tuned on instinct; delinquency found months late
Fraud / FCU team	Edited PDFs and AI-generated statements pass simple checks; circular money makes weak borrowers look busy.	Direct fraud losses; first-payment defaults
Borrower	Rejected with no reason; asked to re-upload; password and format problems; data used without clear consent.	Drop-off; complaints; regulatory risk for the lender

Personas

Persona	Goal	Top frustration	What LedgerLens gives them
Credit officer ("Asha", NBFC, 40 files a day)	Decide quickly and correctly	Re-doing the machine's work because she can't trust it	Every number with its reason and confidence; a short review queue of only the items that move the decision
Senior underwriter / credit committee	Approve exceptions and large tickets	Thin case notes; can't see what was overridden	Case view with overrides, reasons and an audit trail
Risk head	Policy that predicts default	No link between score bands and repayment	Policy engine, champion/challenger, band-vs-delinquency reports
Ops / QC analyst	Clear failed and stuck cases	Password PDFs, jumbled pages, unknown formats	Exception queue with a guided fix for each failure type
FCU / fraud analyst	Catch fake statements before disbursal	Forensics spread across tools	One fraud panel: document, arithmetic and behaviour signals with evidence
DSA / sales	Get the file through fast	Rejections for missing or bad documents	Instant upload check: "statement OK / missing March / wrong password"
Borrower	A loan, fast and fair	Repeated document requests	One AA consent or one upload; clear reason if refused (via the lender)
Lender's developer	Integrate once, run at scale	Unclear states, no webhooks, breaking changes	Versioned API, webhooks, sandbox with sample statements

Key use cases

Use case	Statement type	What decides it	Fit for LedgerLens
Personal loan (salaried)	Salary account, 3–6 months	Salary regularity, FOIR, bounces	Core MVP (prototype covers it)
Personal loan (self-employed / gig)	Savings, several banks	Irregular income averaged over time, many small payers	MVP+ (needs gig income rules)
Credit card / BNPL limit	Savings + existing card	Spend capacity, existing card utilisation	Phase 2 (card behaviour rules)
MSME / business loan	Current account, 6–12 months, often several banks	Turnover, GST vs credits, customer concentration, cheque bounces	Phase 2 (separate scorecard)
Home loan	Salary + savings, 6–12 months, often joint	Stable income, own contribution, existing EMIs	Phase 2 (co-applicant support)
NRI borrower	NRE/NRO + foreign remittances	Remittance regularity, currency	Phase 3
Post-disbursal monitoring	AA data refreshed monthly	Salary stops, new loans, bounces	Phase 3

Value proposition

- **For the lender:** cut time to decision from hours to minutes, and make every decision auditable.
- **For the credit officer:** review 3–5 transactions that matter instead of re-checking 300.
- **For the risk head:** change policy without code, and see which policies actually predict repayment.
- **For the borrower:** one consent or one upload, and a reason when refused.

Differentiation in one table

Incumbents optimise for	LedgerLens optimises for
Number of variables (500+)	The 5–10 that decide this case , each explained

Incumbents optimise for	LedgerLens optimises for
Vendor-owned score	Lender-owned policy on top of transparent components
Fraud flag as a yes/no	Fraud flag with evidence a person can verify
Static rules updated by the vendor	Learning from officer corrections (counterparty rules, category fixes)
PDF report for humans, JSON for machines	One case view shared by officer, committee, auditor and API

3. Current market capabilities (table stakes)

Finding: the prototype already covers about **two-thirds of the table-stakes list**. The missing pieces are mostly **enterprise plumbing**: AA fetch, Excel input, roles, LOS connectors and PDF/Excel reports. They are not new ideas, but no lender will buy without them.

Status key: **Done** = works in the prototype [Confirmed] · **Partial** = exists but not production-grade · **Gap** = not built.

#	Capability	What "good" looks like in the market	LedgerLens today	Status
1	Statement upload: digital PDF	All major Indian banks; multi-file	Rule parser + LLM fallback; multi-file; 25 MB / 300 pages limit	Done
2	Scanned PDF / phone photo	OCR with quality score	Tesseract OCR in browser; confidence $\geq 60\%$ per page warning	Partial (server-side OCR needed)
3	Password-protected PDF	Prompt, decrypt, never store password	Done; password kept in memory only	Done
4	Excel / CSV	Bank exports, quoted fields	CSV done; Excel (.xlsx) not read	Partial
5	Account Aggregator fetch	Consent via an AA, FI data in JSON	Not built	Gap

#	Capability	What "good" looks like in the market	LedgerLens today	Status
6	Net-banking fetch	Borrower logs in, statement pulled	Not built (declining relevance, see note)	Gap (skip)
7	Multi-bank, multi-account, multi-month merge	Accounts kept separate, then combined	Grouped by account number; duplicates across files removed	Done
8	Statement types	Savings, salary, joint, current, credit card	Savings, joint, credit card (sign handled); current account read but scored as personal	Partial
9	Data validation	Page order, balance arithmetic, period, duplicates	22 checks in 6 groups, all in JSON output	Done
10	Transaction categorisation	2-level taxonomy, high accuracy	Rules → LLM → manual; L1/L2 + confidence + method; custom CSV taxonomy	Done (accuracy unmeasured)
11	Salary / income detection	Salary vs P2P, self-transfer, refund, loan disbursement	Done; recurring-company heuristic sent to review	Done
12	EMI / obligation detection + FOIR	Per-loan EMIs, FOIR	Loan a/c or NACH-to-lender rule; FOIR with DECLINE above 65%	Done
13	Bounce, penalty, overdraft	Counts, charges, days	Done	Done
14	Cash flow and average balance	Monthly in/out, average and minimum EOD balance	Done (EOD across deposit accounts)	Done

#	Capability	What "good" looks like in the market	LedgeLens today	Status
15	Transaction fraud	Circular, round-tripping, structuring, pass-through	Circular, structuring (cash 40–50k), overnight pass-through, balance breaks	Done
16	Document tamper detection	PDF metadata, fonts, edit history	Not built; only arithmetic catches edits	Gap
17	Credit scoring and rules	Score + decision + reasons	6 components, 1000 points, hard overrides, reasons	Done (uncalibrated)
18	Reports	PDF and Excel for officers, JSON for systems	JSON export; on-screen views; no PDF/Excel	Partial
19	APIs / webhooks	Async jobs, status, webhooks	API v1: users, statements, jobs, results (versioned), review; no webhooks	Partial
20	LOS / LMS integrations	Connectors to common LOS platforms	None	Gap
21	User and role management	Analyst, approver, admin; SSO	Bearer token per user; no roles	Gap
22	Audit logs	Immutable, per case, exportable	Activity log on server (shared); review decisions versioned	Partial (not tamper-proof)

Notes

- **Net-banking scraping is declining [Assumption]:** it asks borrowers for bank credentials, which lenders and banks increasingly avoid now that AA exists. Recommend **not building it**. [Reg/Legal] Confirm with counsel whether credential-based fetch is acceptable to target lenders.
- **Accuracy is the biggest unknown.** The prototype has never been measured on real, labelled statements. Step 1 of the MVP is a labelled test set (see the PRD).

4. Advanced and differentiating capabilities

Finding: of the 14 advanced capabilities, **five are genuinely open in India** and fit LedgerLens's strengths: explainable scoring, a lender-owned policy engine, learning from corrections, an AI underwriting assistant, and AA-based early warning. Document forensics and MSME underwriting are crowded but *required*: LedgerLens must match them, not lead with them.

Market maturity: **Crowded** = several Indian vendors sell it · **Emerging** = one or two, or only global players · **Open** = not seen sold in India [*Approx. where not sourced*].

Capability	What it does	Market maturity (India)	Value to the lender	LedgerLens head start
LLM extraction for unseen formats	Reads a new bank layout without a hand-made template	Emerging	New banks and co-ops work on day one; fewer "unsupported format" rejections	LLM extraction layer built and tested with a mock; not run on real model at scale
Explainable scoring in plain language	Each component, the score and the decision explained with the borrower's own numbers	Open	Faster officer decisions; defensible to auditors and borrowers	Built (① panels, score guide)
Document forensics	PDF metadata, producer software, fonts, overlays, edit increments, AI-generated layouts	Crowded (Perfios, FinBox; Inscribe globally)	Stops fake statements, the most-forged document type	Only balance arithmetic
MSME cash-flow underwriting	Turnover, GST (GSTR-1/ 3B) vs bank credits, customer concentration, seasonality, cheque bounces	Crowded at the top (ScoreMe, Perfios)	Unlocks business loans, the largest ticket sizes	Reads current accounts; no business scorecard

Capability	What it does	Market maturity (India)	Value to the lender	LedgerLens head start
Gig / irregular income	Many small payers, platform payouts (Swiggy, Uber, Urban Company), weekly cycles	Emerging	Serves a fast-growing borrower group bureaus score poorly	Business-income keywords only
Credit card behaviour	Utilisation, minimum-due-only payments, cash withdrawals, revolving balance	Emerging	Early sign of stress before EMIs bounce	Card statements read; no behaviour rules
Name matching across banks	"ROHAN MEHTA" = "ROHAN K MEHTA" = "MEHTA ROHAN"	Common inside vendors, rarely shown	Fewer false "different holder" alerts; real mismatches stand out	Exact match only
Loan-app stacking, gambling, crypto signals	Repayments to many digital lenders; betting and crypto exchanges	Emerging	Catches over-leverage that the bureau misses for days or weeks	Not built
Policy engine (no code)	Lender sets cut-offs, knock-outs, segment rules and versions them	Open as a self-serve product [Approx.]	Risk team changes policy in hours, not vendor sprints	Thresholds are constants in code
Champion / challenger	Run two policies on live traffic, compare outcomes	Open in BSA tools	Evidence-based policy changes	Not built (versioned results help)

Capability	What it does	Market maturity (India)	Value to the lender	LedgerLens head start
Early warning via AA	Refresh data monthly after disbursal; flag salary stop, new loans, bounces	Emerging	Act before the first missed EMI	Not built
AI underwriting assistant	Officer asks "Is the March jump in income a one-off?" and gets an answer with linked transactions	Open in India; Inscribe has fraud agents	Faster, deeper reviews; consistent answers	LLM layer + explanations to build on
Bureau + statement cross-check	EMIs in the statement vs loans on the bureau report	Emerging	Finds undeclared loans and mismatched EMIs	Not built
Learning from corrections	Officer fixes become rules for next time; accuracy tracked per rule	Open as a visible feature	Review effort falls month after month	Built in part (counterparty rules, MANUAL 100% labels)

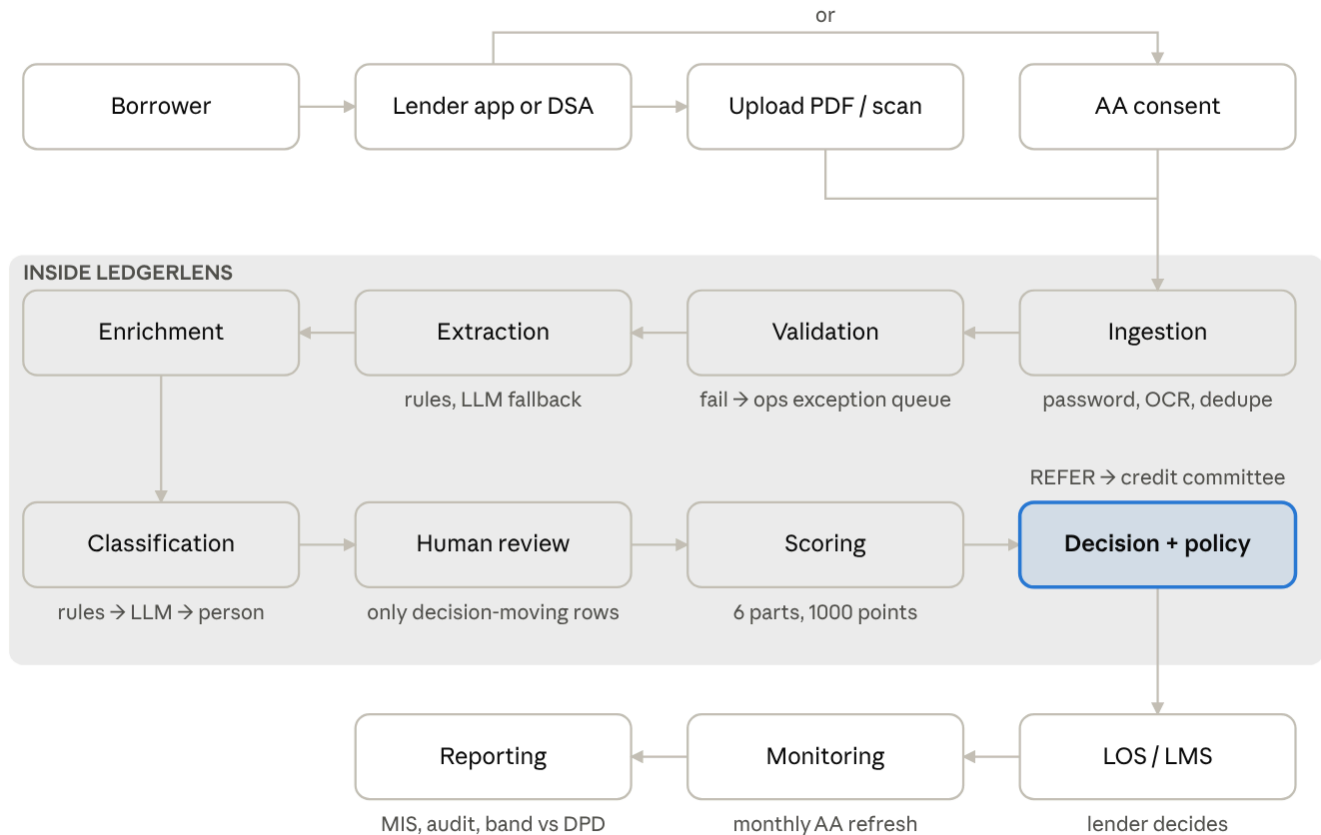
Data-source facts that shape the roadmap

- **GST returns are available through Account Aggregators:** RBI added GSTN as a data provider, sharing GSTR-1 and GSTR-3B with consent ([Sahamati on the ReBIT GSTN schema](#)). MSME underwriting can therefore use one consent for bank and GST data.
- **Bureau data** needs a membership or partnership with a credit information company. **[Reg/Legal]** Confirm the lender (not LedgerLens) pulls the bureau report and passes it in.

5. End-to-end flow

Finding: LedgerLens owns the eight stages in the shaded band. The lender owns capture (its app or DSA) and the final decision (its LOS). Two exits keep bad data out: **failed validation goes to an ops exception queue**, and **REFER goes to a credit committee**.

LedgerLens owns eight stages between borrower data and the lender LOS



end-to-end flow · 4 capture steps, 8 LedgerLens stages, 3 downstream

The flow snakes left to right, then right to left: capture, then extraction, then judgement, then what happens after the decision.

Each stage and the systems involved

#	Stage	What happens	Systems / components	Target time [Assumption]	Prototype today
1	Borrower	Applies for a loan; agrees to share bank data	Lender's app or web form; consent screen	—	n/a
2	Lender app or DSA	Creates a case; chooses upload or AA; sets product (PL, BL, card)	Lender LOS or LedgerLens case API; DSA portal	—	Upload screen only
3	Upload	PDF, scan, photo, CSV/Excel; password asked if needed	Upload API (pre-signed URL), virus scan, file store (encrypted)	< 5 s	Done (no virus scan)

#	Stage	What happens	Systems / components	Target time [Assumption]	Prototype today
3b	AA consent	Borrower approves a consent (purpose, period, frequency) in the AA app; data arrives as JSON	AA gateway (e.g. via a TSP), consent manager, FI-data decryptor	30 s – 2 min	Gap
4	Ingestion	Detects type; decrypts; OCR for scans; splits pages; removes duplicate files	Ingestion service, OCR worker pool, job queue	< 30 s for 20 pages	Done (OCR in browser)
5	Validation	Page order, balance arithmetic, period coverage, holder name, tamper signals; stop or warn	Validation engine (22+ checks), forensics service (Phase 2)	< 5 s	Done (no forensics)
6	Extraction	Header (bank, account, holder, period) and rows (date, narration, debit, credit, balance)	Rule parsers per bank layout; LLM extractor for unknown layouts; reconciler that cross-checks both	< 20 s	Done
7	Enrichment	Channel (UPI, NEFT, NACH, ATM, cheque), counterparty, loan account, IFSC, currency	Narration parser, counterparty normaliser, lender/merchant dictionary	< 5 s	Done
8	Classification	Category L1/L2, confidence, method; income vs transfer; EMI detection	Rule engine, LLM classifier, lender's custom taxonomy and counterparty rules	< 20 s	Done
9	Human review	Only low-confidence rows that can change the decision; fixes become rules	Review queue, assignment, SLA timers	Minutes (async)	Done (no assignment/SLA)
10	Scoring	Six components → 1000-point score and band; segment scorecard	Scoring service (versioned), feature store	< 2 s	Done (one scorecard)

#	Stage	What happens	Systems / components	Target time [Assumption]	Prototype today
11	Decision + policy	Lender's knock-outs and cut-offs on top of the score; APPROVE / CONDITIONS / REFER / DECLINE with reasons	Policy engine (versioned rules), explanation generator	< 1 s	Partial (rules in code)
12	LOS / LMS	Result pushed to the lender; lender makes and records the final credit decision	Webhooks, LOS connectors, result API (versioned JSON + PDF report)	< 5 s after decision	API only, no webhooks
13	Monitoring	After disbursal, AA data refreshed (e.g. monthly); early-warning signals	AA periodic consent, monitoring jobs, alert service	Monthly	Gap
14	Reporting	Portfolio MIS, audit trail, accuracy, score band vs repayment (DPD)	Analytics warehouse, BI dashboards, audit store	Daily	Activity log only

[Reg/Legal] Stages 3b and 13 rely on AA consent with a stated purpose and period; monitoring after disbursal needs its own consent purpose. Stage 11: the regulated lender must stay the decision-maker; LedgerLens provides a recommendation.

6. Product modules and architecture

Finding: twelve product modules sit on one shared platform: one job queue, one data store, one audit log, and one LLM gateway that masks personal data. The two highlighted modules, **Review queue** and **Policy engine**, are where LedgerLens is built to differ. The rest must simply be solid.

Twelve modules share one platform; review and policy set LedgerLens apart



platform architecture · 5 layers, 12 product modules

Read top to bottom: people and systems enter through the edge, the modules do the work, the shared platform keeps state and evidence, and external services supply data.

What each module owns

Module	Owns	Key objects	Primary users	Phase
Dashboard	Today's work and portfolio health at a glance	Widgets, saved views	All	MVP

Module	Owns	Key objects	Primary users	Phase
Case management	One loan application = one case; status, owner, SLA, notes	Case, applicant, co-applicant, product	Credit officer, DSA	MVP
Ingestion	Upload, password, OCR, AA fetch, duplicate files	File, page, AA consent, FI data	DSA, ops	MVP (AA in Phase 2)
Validation & integrity	22+ trust checks; stop / warn / pass	Check result, exception	Ops, FCU	MVP
Transactions	Extracted rows with category, confidence, method; filters and export	Account, transaction, counterparty	Credit officer	MVP
Review queue	Only decision-moving, low-confidence rows; assignment, SLA; fixes become rules	Review item, decision, rule	Credit officer, ops	MVP
Risk scoring & decision	Components, score, band, reasons, explanations	Scorecard version, score, reason code	Credit officer, risk head	MVP
Policy / rules engine	Lender's knock-outs and cut-offs per product, versioned, simulated before go-live	Policy, rule, version, simulation	Risk head	MVP (basic) → Phase 2 (no-code)
Fraud & forensics	Document tamper signals, transaction fraud, evidence pack	Fraud signal, evidence	FCU analyst	MVP (transaction) → Phase 2 (document)
Reports & analytics	Case report (PDF/Excel/JSON), MIS, accuracy, band vs repayment	Report, metric	Risk head, management	MVP (case report) → Phase 2 (MIS)
Monitoring	Post-disbursal AA refresh and early-warning alerts	Watchlist, alert	Collections, risk	Phase 3

Module	Owns	Key objects	Primary users	Phase
Integrations	AA gateway, bureau input, GST via AA, LOS connectors	Connector, mapping	Lender developer	Phase 2
Notifications	Email/SMS/in-app/webhook on case events and SLA breaches	Subscription, event	All	MVP (webhooks, in-app)
Users, roles & permissions	Roles, teams, branches, maker-checker, SSO	User, role, team	Admin	MVP (roles) → Phase 2 (SSO)
Admin / ops console	Tenant settings, taxonomy, thresholds, data retention, health	Tenant, setting	Admin, LedgerLens ops	MVP
Developer / API portal	Docs, sandbox with sample statements, keys, webhook logs	API key, sandbox case	Lender developer	MVP (docs + sandbox)

Architecture principles [Assumption]

1. **Multi-tenant with hard isolation:** every row carries a tenant id; per-tenant encryption keys; no cross-tenant learning without written consent.
2. **Every number is traceable:** each score component links to the transactions that produced it, so explanations and audits come from the same data.
3. **Versioned everything:** parser, taxonomy, scorecard and policy versions are stored on each result, so any past decision can be replayed exactly.
4. **LLM behind a gateway:** personal data masked before any model call; prompts and outputs logged; per-tenant switch to turn LLM use off. **[Reg/Legal]** see the register.

7. Dashboard

Finding: two dashboards, not one. A credit officer needs **"what do I do next"**; a risk head needs **"is the portfolio and the machine healthy"**. Mixing them buries the action items under charts.

Officer dashboard (default for credit officers and ops)

Top to bottom:

1. **Action bar:** my cases waiting (count), review items assigned to me, SLA breaches in the next 2 hours, failed extractions for my branch.
2. **My queue table:** case, applicant, product, amount, stage, decision so far, age, SLA timer; sorted by SLA.
3. **Alerts strip:** fraud flags raised today on my cases, statements with failed validation, password still needed.

Risk and operations dashboard (risk head, ops manager, management)

Layout: a KPI row of six tiles, then four chart panels in a 2 × 2 grid, then the alerts list.

Widget	Definition	Visual	Why it matters	Alert when [Assumption]
Applications processed	Cases with a completed analysis in the period	KPI tile + daily trend line	Volume and capacity planning	Day volume > 2× the 4-week average
Auto-decision rate	Cases decided with zero human review items ÷ all completed cases	KPI tile + weekly line	Main efficiency lever	Drops > 10 points week on week
Approve / conditions / refer / decline mix	Share of each recommended decision	100% stacked bar per week	Policy drift; sudden spikes in DECLINE	Any share moves > 15 points
Average time to decision	Upload (or AA data received) → recommendation; median and 90th percentile	KPI tile (median, p90)	Borrower experience; SLA to lenders	p90 > 10 minutes
Review-queue backlog and SLA	Open review items; % resolved within SLA; oldest item age	KPI tile + ageing bars (0-1 h, 1-4 h, 4-24 h, 24 h+)	Staffing; hidden delays	Any item > 24 h
Extraction accuracy / OCR failures	Share of statements passing balance arithmetic first time; OCR pages below 60% confidence	KPI tile + by-bank table	Data quality is the base of everything	First-pass rate < 95% for any bank with ≥ 50 statements

Widget	Definition	Visual	Why it matters	Alert when [Assumption]
Fraud flags by type	Count by type: balance break, PDF edit signal, circular, structuring, pass-through, name mismatch	Horizontal bar chart	Fraud trends; FCU workload	Any type doubles week on week
Bank / format coverage	Statements by bank and layout version; share parsed by rules vs LLM vs failed	Table with sparkline	Where to build new parsers; LLM cost	New layout with > 20 failures
Income band and FOIR distribution	Applicants by monthly income band and FOIR band	Two histograms (FOIR with the 65% cut-off line marked)	Who is applying; how policy bites	Share above FOIR cut-off > 25%
Score band vs early delinquency	30+ DPD rate at 3 months on book, by score band (needs lender repayment data)	Bar per band	Proves the score works	Bands do not fall monotonically
Alerts / action items	Everything above that crossed a threshold, with owner and link	List	Nothing waits silently	—

Filters on every panel: date range, product, branch / DSA, bank, segment (salaried, self-employed, MSME), policy version.

Design rules: every number clicks through to the cases behind it; every chart shows its definition on hover; numbers show units (₹, %, minutes).

8. Operations console

Finding: every exception has **one owner, one screen and one way out**. The console is built around queues (failed extraction, review, fraud, escalation), not around reports. Every action requires a reason and lands in the audit trail.

Workflow	Trigger	Who	Steps	Outcome and audit
Search cases and transactions	Officer, auditor or support query	Any role (scoped to their team)	1. Search bar: case id, applicant name, PAN (masked), account last 4, phone, DSA. 2. Filters: date, product, stage, decision, bank, flag type. 3. Transaction search inside a case: narration text, amount range, category, counterparty.	Results show stage and decision; every view of personal data is logged
Fix a failed or partial extraction	Validation stop (balance break on many rows, no rows found, unknown layout)	Ops / QC	1. Open exception: shows page image beside extracted rows, failing rows highlighted. 2. Choose fix: re-run with LLM extractor, set column mapping for this layout, or correct header fields (opening balance, account number). 3. Re-validate. 4. If still failing, mark "request new statement" (message goes to DSA).	Case resumes or goes back to DSA; fix saved as a layout hint for the bank

Workflow	Trigger	Who	Steps	Outcome and audit
Review low-confidence transactions	Item in review queue (confidence below threshold and can move the decision)	Credit officer	1. Queue sorted by impact (₹ effect on income or FOIR). 2. Each item: narration, amount, suggested category, why unsure, similar past decisions. 3. Accept, change category or counterparty, or "apply to all from this counterparty". 4. Score re-runs instantly; the change in score is shown.	Item closed as MANUAL 100%; optional rule saved for the tenant
Override a category or decision	Officer disagrees with the recommendation	Credit officer (category); senior underwriter (decision)	1. Click Override. 2. Pick reason from a list (e.g. "income verified by salary slip", "one-off medical expense") + free text. 3. Decision overrides above the officer's authority need a checker.	Original and new value, reason, user and time stored; override rate reported per officer

Workflow	Trigger	Who	Steps	Outcome and audit
Investigate a fraud flag	Flag raised: balance break, PDF edit signal, circular funds, structuring, name mismatch	FCU analyst	1. Fraud panel lists signals with evidence (page crops, metadata, linked transactions shown as a money-flow chain). 2. Compare against an AA pull or a fresh bank-generated statement if available. 3. Verdict: genuine, suspicious (refer), confirmed fraud (decline + report).	Verdict and evidence pack stored; confirmed patterns added to the fraud library
Password-protected statement	PDF needs a password	DSA or ops	1. Case pauses at "Waiting for password". 2. DSA enters it (common bank formats shown as hints, e.g. date of birth). 3. Wrong twice → ask borrower via link.	Password used in memory only, never stored or logged

Workflow	Trigger	Who	Steps	Outcome and audit
Jumbled or missing pages	Page-order check fails	Ops	1. Thumbnails in detected order with page numbers. 2. Drag to reorder, or mark missing. 3. Missing pages → request from DSA; cannot score until complete.	Reorder recorded; scoring blocked while pages are missing
Escalate to credit committee	Decision REFER, override above authority, or ticket above a limit	Senior underwriter	1. "Send to committee" builds a case summary: score, reasons, overrides, fraud verdict, officer note. 2. Committee members approve or decline with conditions in the console, or the summary is exported to the lender's committee process.	Committee decision, members and date recorded

Workflow	Trigger	Who	Steps	Outcome and audit
View audit trail	Audit, complaint, dispute, model review	Auditor, compliance	1. Case timeline: every event (upload, checks, extraction version, review edits, overrides, decisions, exports). 2. "Replay" shows the result exactly as it was at any version. 3. Export as PDF or JSON.	Read-only; append-only storage; export itself logged

Maker-checker [Assumption]: decision overrides, policy changes and taxonomy changes need a second person. Category fixes on a single case do not.

9. Validation and integrity engine

Finding: trust is decided in **three layers**: is the file genuine (document), does the data add up (arithmetic), and does it belong to this applicant for the right period (coverage and identity). Each check returns a **severity**, and the severity alone decides what happens next. Officers never have to interpret a raw check result.

Severities

Severity	Meaning	What happens
STOP	Cannot analyse safely	Case paused; goes to ops exception queue
REFER	Analysis runs, but a person must decide	Decision forced to REFER; FCU or senior review
WARN	Analysis runs; shown to officer	Banner on case; counted in data-quality score
INFO	Recorded only	In report and JSON

The checks

Layer	Check	How it works	Severity	Prototype
Document	Encryption / password	Detect encrypted PDF; ask for password	STOP until provided	Done
Document	File sanity	Type, size ≤ 25 MB, pages ≤ 300, not empty, not duplicate of another file (hash)	STOP / INFO	Done
Document	PDF metadata	Producer and creator software (e.g. a photo editor or online PDF editor instead of the bank's generator), creation vs modification date, author field	REFER if editor detected	Gap
Document	Incremental edits	PDF saved again after creation (extra revision sections), objects added on top of text (overlays, white boxes)	REFER	Gap
Document	Font and layout fingerprint	Fonts, sizes, spacing and logo position compared with known genuine samples of that bank's layout	REFER on mismatch	Gap
Document	Text layer vs image	For PDFs with both, OCR of the image must match the text layer	REFER on mismatch	Gap
Document	Digital signature	Validate bank's digital signature where present	INFO (pass) / REFER (broken)	Gap

Layer	Check	How it works	Severity	Prototype
Document	AI-generated / template fakes	Signs of a synthetic statement (too-regular spacing, repeated narrations, impossible reference numbers)	REFER	Gap (Phase 2+)
Arithmetic	Running balance	Previous balance $\pm$ amount = printed balance, every row	REFER if any break	Done
Arithmetic	Opening + flows = closing	Per account and period	REFER	Done
Arithmetic	Valid amounts and dates	One of debit/credit per row; dates between 2000 and today; in statement period	WARN (rows dropped or flagged)	Done
Arithmetic	Reference-number sanity	UPI/IMPS reference formats and uniqueness; cheque-number sequence	WARN	Gap
Coverage	Page order and completeness	Page X of Y footers; date continuity when no footers	STOP	Done
Coverage	Period coverage	Data covers the stated period; no gap > 35 days; ≥ 3 months and ≥ 20 transactions (configurable by product)	WARN (STOP if policy requires 6 months)	Done
Coverage	Duplicates across files	Same transaction in overlapping uploads removed once	INFO	Done
Identity	Holder name consistency	Fuzzy match across accounts (initials, order, spacing); joint holders allowed	WARN if weak match; REFER if clearly different person	Partial (exact match only)

Layer	Check	How it works	Severity	Prototype
Identity	Name vs application	Statement holder vs applicant name and PAN name from the LOS	REFER on mismatch	Gap
Identity	Account ownership	Account number vs the lender's penny-drop or AA-linked account	INFO / REFER	Gap
Cross-source	PDF vs AA	If both exist for the same account and period, balances and rows must match	REFER on mismatch (strong fraud signal)	Gap

Exception handling and manual verification

1. **STOP** creates an exception in the ops queue with the failing pages and rows highlighted (workflow in section 8).
2. **REFER** opens the fraud panel with the evidence for each signal; the analyst records genuine / suspicious / fraud.
3. **Manual verification options** offered in the panel: request an AA pull, request a statement emailed directly by the bank, or call the branch. The chosen method and outcome are stored.
4. A check cannot be "switched off" on a case. It can only be **overridden with a reason** by a role allowed to do so.

Validation report

Every case carries a **trust summary**: overall status (Trusted / Trusted with warnings / Needs review / Failed), each check with result and evidence, and the parser, OCR and check versions used. It is part of the case report (PDF/Excel) and the JSON (`data_validation` , already present in the prototype).

[Reg/Legal] Confirm with counsel how a confirmed forged statement must be reported (lender's fraud reporting process) and how long evidence is kept.

10. Risk and decision engine

Finding: split the engine into **three layers the lender can see**. The *scorecard* measures (six components, 0-1000). The *policy* decides (the lender's knock-outs and cut-offs). The *explanation* tells the officer why, with the borrower's own numbers. Today the prototype mixes score and policy in code; separating them is the single most important architecture change.

Layer 1: scorecard components [Confirmed in prototype]

Component	Weight (salaried)	What it measures	Key metrics
Income stability	25%	Is income regular and enough?	Regularity (variation of main source), number of sources, growth first → last month, level
Debt service	20%	Can they afford existing EMIs?	FOIR (EMIs ÷ income), EMI bounce rate, on-time rate
Liquidity	15%	Is there a cushion?	Average and minimum end-of-day balance vs income and EMI; negative-balance days
Banking behaviour	10%	Do they handle the account responsibly?	Bounces, penalty fees, overdraft days, charges
Fraud indicators	15%	Is anything suspicious?	Balance breaks, circular money, overnight pass-through, structuring
Expense management	15%	Is spending under control?	Discretionary share, month-on-month spend trend, savings rate after EMIs

Composite = $\Sigma(\text{component score} \times \text{weight}) \times 10$. Bands: Excellent ≥ 800 · Good 700–799 · Fair 600–699 · Below average 500–599 · Poor < 500 .

Segment scorecards [Assumption: proposed weights, to be calibrated]

Segment	Income	Debt service	Liquidity	Banking	Fraud	Expense	What changes in the metrics
Salaried	25%	20%	15%	10%	15%	15%	As today

Segment	Income	Debt service	Liquidity	Banking	Fraud	Expense	What changes in the metrics
Self-employed / gig	20%	20%	20%	10%	15%	15%	Income averaged over 6–12 months; many small payers count as stability, not noise; platform payouts recognised
MSME (current account)	25% (turnover)	20% (DSCR)	15%	15%	15%	10% (margin proxy)	Turnover = business credits net of own transfers and loans; GST-declared turnover vs bank credits; top-5 customer share; seasonality; cheque bounces weigh more
Credit card / BNPL limit	20%	25%	15%	10%	15%	15%	Adds card utilisation, minimum-due-only months, cash withdrawals on card

Layer 2: hard overrides and policy

Hard overrides already in the prototype **[Confirmed]**: no income → DECLINE; FOIR > 65% → DECLINE; balance break, structuring or circular money → REFER; any EMI bounce → at most APPROVE WITH CONDITIONS.

Policy layer to build **[Assumption]**, set per lender and per product, versioned, with maker-checker:

Rule type	Example
Knock-out	Statement older than 45 days → STOP; fewer than 6 months for business loans → STOP
Cut-off	Personal loan: score ≥ 650 and FOIR ≤ 50% → APPROVE
Band mapping	Score 600–649 → APPROVE WITH CONDITIONS (lower amount, co-applicant)
Amount rule	Maximum EMI = 50% of average monthly income – existing EMIs
Segment routing	Self-employed with GST data → MSME scorecard

Rule type	Example
Referral trigger	Ticket > ₹10 lakh → REFER to committee regardless of score

Simulation before go-live: a new policy version is run on the last 90 days of cases. The risk head sees how many decisions would change before publishing.

Layer 3: explanation and AI assistance

- **Explanation [Confirmed in prototype]:** each component, the score and the decision have an ⓘ panel with "what it measures / how it is scored / this statement", for example "₹61,750 a month ÷ income ₹1,67,111 = FOIR 37% → 85/100".
- **Reason codes [Assumption]:** each decision carries 1–4 reason codes (e.g. R12 "FOIR above policy") for the LOS and for adverse-action letters.
- **AI-assisted, never AI-decided:** the LLM is used for reading (extraction, unclear narrations, officer Q&A). Scores and decisions come only from versioned, deterministic rules and models. **[Reg/Legal]** Confirm this boundary satisfies the lender's model-risk policy and RBI expectations on explainability (see register).
- **Statistical model later (Phase 3):** once there is repayment data, a gradient-boosted model can be added as a *challenger* to the rules scorecard, with reason codes from feature contributions.

Manual review and overrides

Officers can change categories (score re-runs) and propose a different decision with a reason. Decision overrides above their authority go to a checker. Override rates per officer, reason and policy rule are reported monthly. A high override rate on one rule means the rule is wrong, not the officers.

Calibration

The prototype's weights and cut-offs are **expert judgement, uncalibrated**. Calibration needs **6–12 months of repayment outcomes** for scored loans. Target: delinquency (30+ DPD at 6 months on book) falls steadily from the lowest to the highest band.

11. API and technical architecture

Finding: the prototype's API v1 already has the right shape: asynchronous jobs, versioned results, review decisions. Production needs four additions: **cases as the top-level object, webhooks, idempotency keys, and enterprise auth**. Data must stay in India, with personal data masked before any LLM call.

Endpoints (v1, production) [Assumption: proposed]

Area	Endpoint	Purpose	In prototype today
Cases	POST /v1/cases	Create a case (applicant ref, product, segment, LOS id)	No (jobs only)
Cases	GET /v1/cases/{id}	Status, stage, latest result summary	GET /jobs/:id
Documents	POST /v1/cases/{id}/documents	Get a pre-signed upload URL; register file	POST /statements
Documents	POST /v1/cases/{id}/documents/{doc}/password	Supply a PDF password (memory only)	POST /jobs/:id/password
AA	POST /v1/cases/{id}/aa-consents	Start an AA consent request (purpose, FI types, period, frequency)	No
AA	GET /v1/cases/{id}/aa-consents/{cid}	Consent status: pending, active, rejected, revoked, expired	No
Analysis	POST /v1/cases/{id}/analyses	Run (or re-run) analysis with a policy version	POST /jobs
Results	GET /v1/cases/{id}/result?version=	Full JSON: accounts, transactions, validation, components, score, decision, reasons	GET /jobs/:id/result? version
Results	GET /v1/cases/{id}/report? format=pdf xlsx	Human-readable case report	No
Results	GET /v1/cases/{id}/transactions? category=&page=	Paged transactions	Inside result
Review	GET /v1/cases/{id}/review-items	Open review items	GET /jobs/:id/review

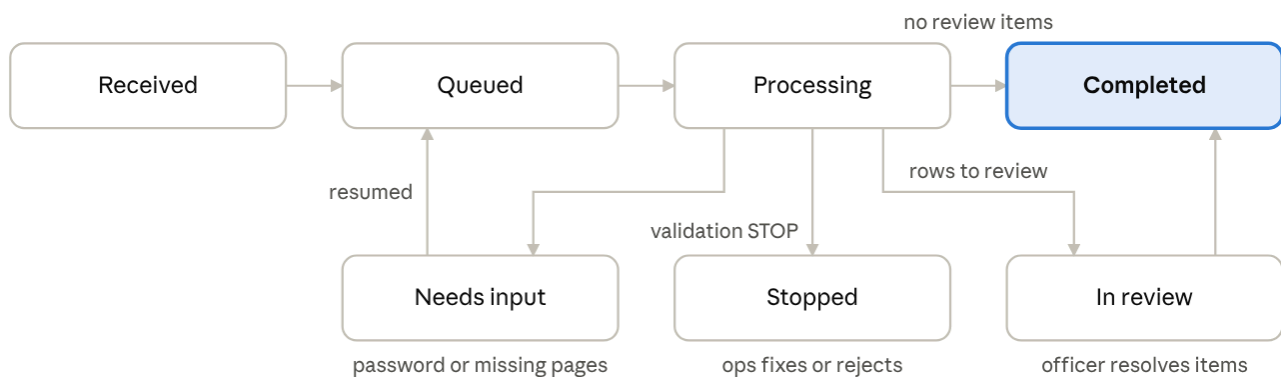
Area	Endpoint	Purpose	In prototype today
Review	<code>POST /v1/cases/{id}/review-items/{item}/resolve</code>	Category / counterparty fix; creates a new result version	<code>POST /jobs/:id/review</code>
Decision	<code>POST /v1/cases/{id}/overrides</code>	Decision override with reason (maker-checker)	No
Policy	<code>POST /v1/policies</code> , <code>POST /v1/policies/{id}/simulate</code>	Create and simulate policy versions	No
Ops	<code>GET /v1/queue</code> , <code>GET /v1/health</code>	Queue depth, worker status	Yes

Webhooks

Events: `case.needs_input`, `case.stopped`, `case.in_review`, `case.completed`, `case.result_updated`, `aa.consent_updated`, `monitoring.alert`. Each delivery is signed (HMAC-SHA256 with a per-tenant secret and a timestamp), retried with exponential backoff for 24 hours, and visible in a delivery log in the developer portal. Payloads carry ids and status only; the lender fetches details with its credentials.

Job states

Every case ends Completed or Stopped; review and missing input only pause it



case / job states · 7 states

Any state can also end in **Failed** (after two automatic retries; the prototype already retries and re-queues on restart). A new review fix or policy run creates a **new result version**; old versions stay readable.

Cross-cutting design

Concern	Design [Assumption unless marked]
Authentication	Server-to-server: OAuth 2.0 client credentials + optional mTLS. Console users: SSO (SAML / OIDC) with MFA. Prototype uses bearer tokens stored as SHA-256 hashes [Confirmed] .
Authorisation	Roles (officer, senior underwriter, ops, FCU, risk head, admin, auditor) × scopes (team, branch, product); tenant isolation on every query
Idempotency	Idempotency-Key header on every POST; key + body hash stored 24 h; replay returns the original response
Errors	RFC 9457 problem+json: type, title, status, code (e.g. PASSWORD_REQUIRED, PAGES_MISSING, UNSUPPORTED_FILE), detail, case_id, trace_id
PII handling	Encryption at rest (per-tenant keys in a KMS) and in transit (TLS 1.2+); account numbers and PAN masked in UI and logs; field-level encryption for names and phone; raw files kept for a configurable period (e.g. 90 days), then deleted with a deletion log. [Reg/Legal] retention periods.
Data residency	All storage and processing in Indian cloud regions. [Reg/Legal] confirm whether any LLM call may leave India.
LLM controls	LLM gateway masks names, account numbers, PAN, phone, UPI ids before the call; no-training contract with provider; per-tenant on/off switch; prompt and response logged with case id; daily cost cap; falls back to rules if the model is slow or down (prototype already labels fallback rows RULE [Confirmed]).
Monitoring	Metrics per stage (latency, error rate, queue depth, LLM tokens and cost), traces across services (OpenTelemetry), SLOs: API 99.9% monthly, p90 time to decision < 10 min

Concern	Design [Assumption unless marked]
Logging	Structured JSON logs without PII; separate append-only audit store for user actions and decisions (hash-chained so edits are detectable)
Scaling	Stateless API behind a load balancer; managed Postgres (prototype uses SQLite [Confirmed]) with read replicas; object storage for files; a managed queue with autoscaled worker pools for OCR and LLM; fair scheduling across tenants (prototype already schedules fairly across users [Confirmed])

12. User journeys

Finding: across all nine journeys the same three moments decide the experience: **the first 30 seconds after upload** (is the file usable?), **the review queue** (only what matters), and **the decision screen** (why, in plain words). Each journey below ends with what "good" looks like.

J1. Credit officer analyses a salaried applicant

1. Asha opens the case from her queue; the statement arrived from the DSA 4 minutes ago and is already analysed.
2. Trust summary: **Trusted**, 3 months, 1 account, holder name matches the application.
3. Decision card: **742 / 1000, Good → APPROVE**. Top reasons: salary ₹1.4 lakh credited on the 1st every month; FOIR 28%; no bounces.
4. She opens ⓘ on Debt service to confirm FOIR: ₹39,200 of EMIs ÷ ₹1,40,000 income.
5. Review queue for this case: empty (all low-confidence rows were small and auto-accepted).
6. She clicks **Accept recommendation**; result and report go to the LOS by webhook.

Good: decision in under 3 minutes with zero manual re-checking.

J2. Underwriting an MSME current account

1. DSA uploads 12 months of a current account (two banks) and the borrower gives AA consent for GST returns.
2. Segment router picks the **MSME scorecard** because the account type is Current and GST data is present.

3. Business view: monthly turnover, GST-declared turnover vs bank credits (e.g. 94% match), top-5 customers = 61% of credits, seasonality (Oct-Dec peak), inward cheque bounces.
4. Officer reviews 4 flagged credits: two large ones from the promoter's personal account are re-labelled **capital infusion, not turnover**.
5. Score re-runs; DSCR drops from 1.6 to 1.3; decision stays APPROVE WITH CONDITIONS (lower limit).

Good: turnover the officer can defend, reconciled to GST, in one screen.

J3. Applicant with several banks

1. Upload: HDFC salary account, SBI savings, ICICI credit card, in one go.
2. Ingestion groups three accounts; holder names "ROHAN MEHTA", "ROHAN K MEHTA", "MEHTA ROHAN" are **fuzzy-matched as the same person** (WARN-free).
3. Transfers HDFC → SBI are tagged **self-transfer, not income**; the card bill paid from HDFC is a transfer, card spends counted once.
4. Coverage check: SBI has only 2 of 3 months → WARN with "request March statement" button.

Good: one combined income and FOIR, no double counting, and a clear note on what is missing.

J4. Password-protected and scanned statement

1. DSA uploads a password-protected PDF and a phone photo of a passbook page.
2. Case pauses at **Needs input: password**; DSA sees the bank's usual password format as a hint and enters it.
3. The photo goes to OCR; page confidence 71% (above 60%) → continue, with a WARN that two amounts were read with low confidence.
4. Balance arithmetic passes, which confirms the OCR digits.

Good: no back-and-forth email; the officer sees exactly which numbers came from OCR.

J5. Tampered statement detected

1. Validation finds 3 running-balance breaks in May and PDF metadata showing an online PDF editor saved the file after the bank generated it.
2. Decision forced to **REFER**; case appears in the FCU queue with evidence: the three rows, the expected vs printed balance, the metadata.
3. FCU requests an AA pull for the same account and month. AA data shows salary of ₹52,000, not the ₹82,000 printed.

4. Verdict: **confirmed fraud**; decision DECLINE; pattern (editor + salary inflation) added to the fraud library.

Good: fraud caught before disbursal, with an evidence pack that stands up to audit.

J6. Low-confidence transactions in review

1. Case has 7 low-confidence rows; only **2 can change the decision** (both large monthly credits with no salary keyword).
2. Review item shows: "₹65,000 from SHARMA ENTERPRISES on the 3rd of each month; suggested Salary (78%) because it repeats from a company."
3. Officer confirms Salary and ticks "apply to this counterparty" → a rule is saved for the lender.
4. Score rises from 655 to 712; decision moves from CONDITIONS to APPROVE; the change is shown before she closes the case.

Good: 2 decisions instead of 7, and next month this employer is recognised automatically.

J7. Override and credit-committee escalation

1. Score 590 (REFER) because of high spending in one month; the applicant had a documented hospital bill.
2. Officer proposes **APPROVE WITH CONDITIONS**, reason "one-off medical expense, receipts verified".
3. The override is above her authority, so it goes to the senior underwriter, who sends it to committee with the auto-built case summary.
4. Committee approves with a lower amount; decision, members and conditions are recorded.

Good: exceptions are fast and traceable; override statistics show whether this rule needs tuning.

J8. AA-based fetch with consent

1. Borrower chooses "Share via Account Aggregator" in the lender's app; consent shows purpose (loan underwriting), data (bank statements, 12 months), frequency (once) and expiry.
2. Borrower approves in the AA app with OTP; data arrives as structured JSON in about a minute.
3. LedgerLens skips OCR and most document checks; integrity is high because the data comes from the bank.
4. Analysis completes; the borrower's lender app shows "documents received".

Good: no PDF, no password, no tamper risk. **[Reg/Legal]** consent purpose and retention must match the AA framework.

J9. Post-disbursal early-warning alert

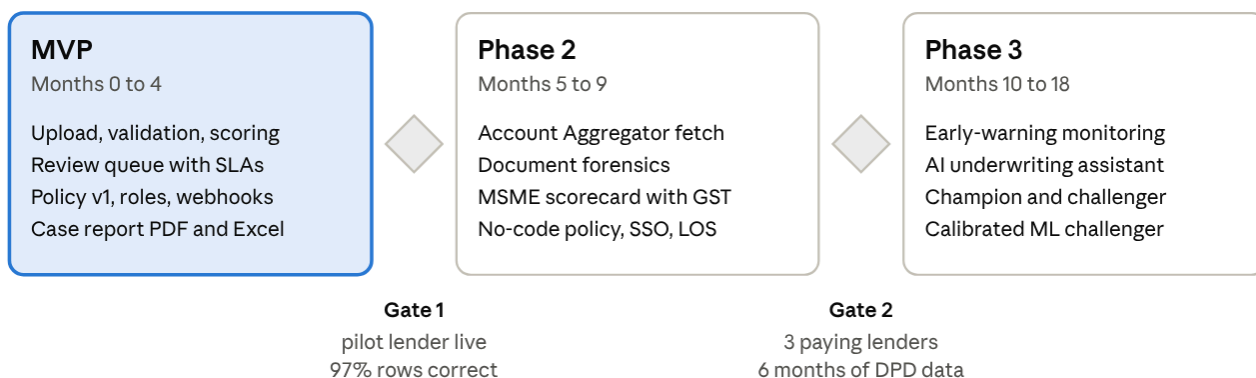
1. With a separate monitoring consent, the AA data refreshes monthly for 12 months.
2. Month 5: salary credit missing, two new loan-app EMIs, one NACH bounce.
3. Monitoring raises a **High** alert to the collections team with the three signals and their dates.
4. Collections calls the borrower before the next EMI date and agrees a restructured plan.

Good: the lender acts weeks before the first missed EMI. **[Reg/Legal]** monitoring must be a disclosed, consented purpose.

13. Roadmap and prioritisation (MoSCoW)

Finding: the MVP is the prototype hardened for one pilot lender: production storage, roles, webhooks, a case report and a measured accuracy baseline. **Account Aggregator and document forensics come in Phase 2**, when there are paying lenders to justify the integrations. Intelligence features need repayment data, so they wait for Phase 3.

Ship the reviewer-first core in 4 months, then data rails, then intelligence



roadmap · 3 phases, 2 gates · months are an assumption

Phases are equal-width bands, not to scale. Month ranges are **[Assumption]** for a team of about 6 engineers, 1 designer and 1 PM.

Feature prioritisation

MoSCoW is set **per phase**: Must = the phase fails without it; Should = high value, can slip one phase; Could = nice to have; Won't = explicitly out of this phase.

Feature	Phase	MoSCoW	Why it belongs here	Prototype
PDF, scan, password, CSV upload	MVP	Must	The core input; lenders will not pilot without it	Done
Excel (.xlsx) upload	MVP	Should	Common bank export; small effort	Gap
22 validation checks + trust summary	MVP	Must	Trust is the product's promise	Done
Fuzzy holder-name matching	MVP	Must	Exact match causes false alarms on every multi-bank case	Gap
Categorisation rules → LLM → review	MVP	Must	Core engine	Done
Review queue with assignment and SLA	MVP	Must	The differentiator must work at pilot scale	Partial
Learning from corrections (counterparty rules)	MVP	Should	Shows review effort falling; cheap since partly built	Partial
Six-component score + hard overrides	MVP	Must	Core output	Done
① explanations + reason codes	MVP	Must	Differentiator; reason codes needed by LOS	Explanations done; codes gap
Policy engine v1 (config file per tenant, versioned)	MVP	Must	Lenders' cut-offs differ; code changes per lender do not scale	Gap
Cases, roles, maker-checker for overrides	MVP	Must	Enterprise basics; audit	Gap
Webhooks + idempotency + problem+json errors	MVP	Must	LOS integration needs push, not polling	Gap
Case report PDF / Excel	MVP	Must	Credit committee and audit use paper-like reports	Gap

Feature	Phase	MoSCoW	Why it belongs here	Prototype
Postgres, object storage, India region, encryption	MVP	Must	Prototype's SQLite on one server is not production	Gap
Labelled test set + accuracy dashboard	MVP	Must	Gate 1 needs a measured baseline	Gap
Server-side OCR	MVP	Should	Browser OCR is slow on large scans	Gap
Developer portal + sandbox	MVP	Should	Speeds pilot integration	Partial (API docs in README)
Account Aggregator fetch	Phase 2	Must	Clean, tamper-proof data; where the market is moving	Gap
Document forensics (metadata, edits, fonts)	Phase 2	Must	Most-forged document; incumbents already have it	Gap
MSME scorecard + GST via AA	Phase 2	Must	Largest ticket sizes; needs AA first	Gap
No-code policy editor + simulation	Phase 2	Must	Turns policy v1 into a self-serve product	Gap
SSO, branch / team scopes	Phase 2	Must	Required by larger NBFCs and banks	Gap
LOS connectors (top 2-3 platforms)	Phase 2	Should	Cuts integration time	Gap
Credit card behaviour rules	Phase 2	Should	Card and BNPL use cases	Gap
Gig income rules	Phase 2	Should	Growing segment	Gap
Loan-app stacking, gambling, crypto signals	Phase 2	Should	Cheap rules on existing categories; real risk signal	Gap
Bureau cross-check (lender passes report)	Phase 2	Could	Useful; depends on lender integration	Gap

Feature	Phase	MoSCoW	Why it belongs here	Prototype
Net-banking credential fetch	Any	Won't	Declining, credential risk; AA replaces it	n/a
Post-disbursal monitoring via AA	Phase 3	Must	Needs AA and live portfolios	Gap
AI underwriting assistant (Q&A on a case)	Phase 3	Should	Needs mature data model and LLM controls	Gap
Champion / challenger	Phase 3	Should	Needs volume and outcomes	Gap
Calibrated statistical model (challenger)	Phase 3	Could	Needs 6–12 months of repayment data	Gap
NRI / foreign-currency scorecard	Phase 3	Could	Niche segment	Partial (currency detected)
Fully automated decisions without officer	Any	Won't (for now)	[Reg/Legal] and trust; keep a human accountable	n/a

14. MVP PRD

Summary: take the LedgerLens prototype to a production pilot with one NBFC for salaried and self-employed personal loans. The pilot runs in parallel with the lender's current process for 8 weeks; success is a measured accuracy baseline and a 50% cut in officer time per file.

Problem statement

Credit officers at NBFCs spend 20–45 minutes per file *[Approx.]* re-checking bank-statement reports they do not trust: salary, EMIs, bounces and possible tampering. Reports are long, unexplained and vendor-scored, so decisions vary between officers and are hard to defend in audits. Lenders need a statement analysis they can trust, explain and tune themselves.

Goals (pilot, 8 weeks)

#	Goal	Measure	Target [Assumption]
G1	Officers decide faster	Median officer handling time per case (open → accept)	≤ 5 minutes (from baseline 20+)
G2	Data is right	Rows with correct date, amount and balance on the labelled set	≥ 97%
G3	Categories are right where it matters	Accuracy on income, EMI and transfer categories	≥ 95%
G4	Review stays small	Median review items per case	≤ 3
G5	Recommendations agree with good judgement	Agreement between LedgerLens decision and lender's final decision	≥ 85%, with every disagreement explained
G6	Reliable	API availability; p90 time to recommendation	99.5%; ≤ 10 minutes

Non-goals (MVP)

- Account Aggregator fetch, document forensics, MSME scorecard (Phase 2).
- Automated final decisions without an officer.
- Net-banking credential fetch (never).
- A calibrated statistical model (needs repayment data).
- Mobile app for borrowers (the lender's app handles capture).

Personas in scope

Credit officer (primary), senior underwriter (overrides), ops/QC (exceptions), risk head (policy v1 and reports), lender developer (API). Details in section 2.

User stories

ID	As a...	I want to...	So that...	Priority
US1	Lender developer	create a case and upload statements by API and get a webhook when done	our LOS gets results without polling	Must
US2	DSA / ops	be told within 30 seconds if a statement is unusable (password, missing pages, wrong file)	I fix it while the borrower is still with me	Must
US3	Credit officer	see a trust summary, score, decision and top reasons on one screen	I decide in minutes	Must
US4	Credit officer	open ⓘ on any number and see how it was calculated from this statement	I can trust and defend it	Must
US5	Credit officer	review only the transactions that can change the decision, sorted by impact	I don't re-check 300 rows	Must
US6	Credit officer	save a fix as a rule for this counterparty	the same item is right next time	Should
US7	Senior underwriter	approve or reject decision overrides with a reason	exceptions are controlled and auditable	Must
US8	Risk head	set our cut-offs and knock-outs per product in a versioned config	policy matches our credit policy, not the vendor's	Must
US9	Ops	fix a failed extraction with the page image beside the rows	cases don't get stuck	Must
US10	Auditor	replay any past decision exactly as it was	audits and complaints are answered with evidence	Must
US11	Credit officer	download a PDF/Excel case report	the credit committee file is complete	Must
US12	Risk head	see accuracy, review load and decision mix by week	I know whether the tool is working	Should

Functional requirements

ID	Requirement	Stories
FR1	Cases API: create, get, list; case holds applicant reference, product, segment, LOS id	US1
FR2	Document upload via pre-signed URL; PDF (digital, scanned, encrypted), JPG/PNG, CSV, XLSX; ≤ 25 MB, ≤ 300 pages	US1, US2
FR3	Pre-check within 30 s: file type, password needed, page order, period, readable text; result shown to uploader and sent by webhook	US2
FR4	Extraction: rule parsers + LLM fallback; header and rows; per-row source (rule / LLM / OCR) and confidence	US3
FR5	All 22 existing validation checks plus fuzzy holder-name match; trust summary with severity	US2, US3
FR6	Categorisation rules → LLM → review with confidence, method and reason per row	US3, US5
FR7	Review queue: only items below threshold and able to move income, FOIR or a hard rule; sorted by ₹ impact; assignment and SLA timer	US5
FR8	Counterparty rule from a review fix, scoped to the tenant, with an undo	US6
FR9	Scoring: six components, 1000 points, bands, hard overrides; results versioned with parser, taxonomy, scorecard and policy versions	US3, US10
FR10	Policy v1: per-tenant, per-product config (cut-offs, knock-outs, band mapping, amount rule), versioned, maker-checker publish	US8
FR11	Explanations: ⓘ for score, decision and each component; 1-4 reason codes per decision	US4
FR12	Overrides: category (officer), decision (officer proposes, checker approves); reason list + free text	US7
FR13	Exceptions queue for STOP cases with page image, highlighted rows, re-run options	US9

ID	Requirement	Stories
FR14	Audit: append-only event log per case; replay by version; export	US10
FR15	Case report: PDF and Excel with summary, trust, components, reasons, transactions, review and override history	US11
FR16	Webhooks (signed, retried 24 h) for needs_input, stopped, in_review, completed, result_updated	US1
FR17	Roles: officer, senior underwriter, ops, risk head, admin, auditor; tenant isolation	US7, US8
FR18	Pilot dashboard: volume, auto-decision rate, review backlog, decision mix, first-pass extraction rate	US12

Non-functional requirements

Area	Requirement
Performance	p90 time to recommendation $\leq$ 10 min for $\leq$ 50 pages; pre-check $\leq$ 30 s; API p95 $\leq$ 500 ms for reads
Scale	5,000 cases a day per tenant at pilot; workers scale horizontally
Availability	99.5% monthly in pilot, 99.9% target at GA; no data loss on restart (jobs re-queued)
Security	TLS 1.2+; encryption at rest with per-tenant keys; MFA for console; OWASP ASVS L2; annual penetration test before bank customers
Privacy	PII masked in UI lists, logs and LLM prompts; raw files deleted after the configured period; no data used to train shared models without written consent
Residency	All storage and processing in India regions [Reg/Legal]
Auditability	Every result reproducible from stored versions; audit log append-only and hash-chained
Accessibility	WCAG 2.1 AA for the console; works on a 1366 px laptop and a phone for DSAs

Acceptance criteria (key examples)

- **AC-US2:** Given a password-protected PDF, when it is uploaded without a password, then within 30 s the case shows "Needs input: password" and a `case.needs_input` webhook is sent; the password is never written to disk or logs.
- **AC-US3:** Given a completed case, when the officer opens it, then score, band, decision, top 3 reasons and trust status are visible without scrolling on a 1366 × 768 screen.
- **AC-US5:** Given 7 low-confidence rows of which 2 can move FOIR or income, then the review queue shows exactly those 2, highest ₹ impact first, and the other 5 are auto-accepted and labelled.
- **AC-US7:** Given an officer proposes a decision override above her authority, then it is not applied until a senior underwriter approves; both actions, reasons and times are in the audit log.

- **AC-US8:** Given the risk head publishes policy v2, then cases analysed before publication keep v1 results, new cases use v2, and the case shows which version decided it.
- **AC-US10:** Given any past case and version, when "Replay" is run, then the score and decision are identical to the stored ones.

Success metrics

Primary: G1 officer handling time and G2 row accuracy. Secondary: G4 review items, G5 agreement, override rate < 10%, pilot lender NPS from officers ≥ 30 .

Dependencies

Dependency	Owner	Needed by
Pilot lender with 300+ historical statements and their final decisions (for the labelled set)	Lender	Week 0
Lender's credit policy for personal loans (cut-offs, knock-outs)	Lender risk team	Week 2
LOS webhook endpoint and test environment	Lender IT	Week 6
LLM provider contract with no-training, India or approved-region processing	LedgerLens + legal	Week 2
Cloud account in an India region; KMS; managed Postgres	LedgerLens	Week 1
Security review / vendor assessment by the lender	Lender InfoSec	Before go-live

Risks

Risk	Likelihood	Impact	Mitigation
Accuracy on real statements lower than on test data	Medium	High	Labelled set first; per-bank accuracy tracking; LLM fallback; ops exception queue
Lender's InfoSec review delays pilot	High	Medium	Start the security questionnaire in week 1; SOC 2 / ISO 27001 roadmap

Risk	Likelihood	Impact	Mitigation
LLM use rejected by lender compliance	Medium	High	Per-tenant LLM switch; rules-only mode already works; masking
Officers don't trust a new score	Medium	High	Run in parallel; explanations; show agreement with their own decisions
Scope creep into AA and MSME	High	Medium	Non-goals agreed with pilot lender in writing
Uncalibrated score causes poor decisions	Low (parallel run)	High	Recommendation only during pilot; lender decides

Assumptions

1. One pilot NBFC, personal loans ₹50,000 to ₹5 lakh, salaried and self-employed.
[Assumption]
2. Statements arrive as PDFs or scans from the lender's DSA flow. [Assumption]
3. The lender keeps the final decision; LedgerLens provides a recommendation.
[Assumption, Reg/Legal]
4. Prototype code is the starting point; storage moves from SQLite to Postgres.
[Confirmed prototype, Assumption for plan]

15. KPIs

North-star metric [Assumption]: officer minutes per decided case. It captures accuracy (bad data means more checking), explanations (trust means less checking) and review design (fewer, better items) in one number the lender's operations head already cares about.

Targets are [Assumption] for the end of the pilot and for 12 months after launch.

Group	KPI	Definition	Pilot target	12-month target
Quality	Extraction accuracy	Rows with correct date, amount, debit/credit and balance ÷ rows on the labelled set	≥ 97%	≥ 99%

Group	KPI	Definition	Pilot target	12-month target
Quality	First-pass validation rate	Statements passing balance arithmetic without manual fixes	≥ 92%	≥ 97%
Quality	Categorisation accuracy	Correct L2 category on labelled rows; reported separately for decision-critical categories (income, EMI, transfer)	≥ 90% overall, ≥ 95% critical	≥ 95% / ≥ 98%
Quality	Bank / format coverage	Share of statements whose layout parses with rules (not LLM, not failed)	≥ 80%	≥ 95%
Efficiency	Auto-decision rate	Cases with zero review items ÷ completed cases	≥ 40%	≥ 65%
Efficiency	Review effort per case	Median review items; median officer minutes per case	≤ 3 items; ≤ 5 min	≤ 2 items; ≤ 3 min
Efficiency	Time to decision	Data received → recommendation, p50 and p90	p90 ≤ 10 min	p90 ≤ 3 min
Trust	Override rate	Decisions changed by officers ÷ decisions; also by reason and by rule	< 10%	< 5%
Trust	Agreement with lender	LedgerLens recommendation = lender's final decision	≥ 85%	≥ 90%
Risk	Fraud catch rate	Confirmed fraudulent statements flagged REFER ÷ all confirmed fraudulent statements	Baseline	≥ 90%
Risk	Fraud false-positive rate	Fraud REFERs found genuine ÷ all fraud REFERs	Baseline	≤ 30%

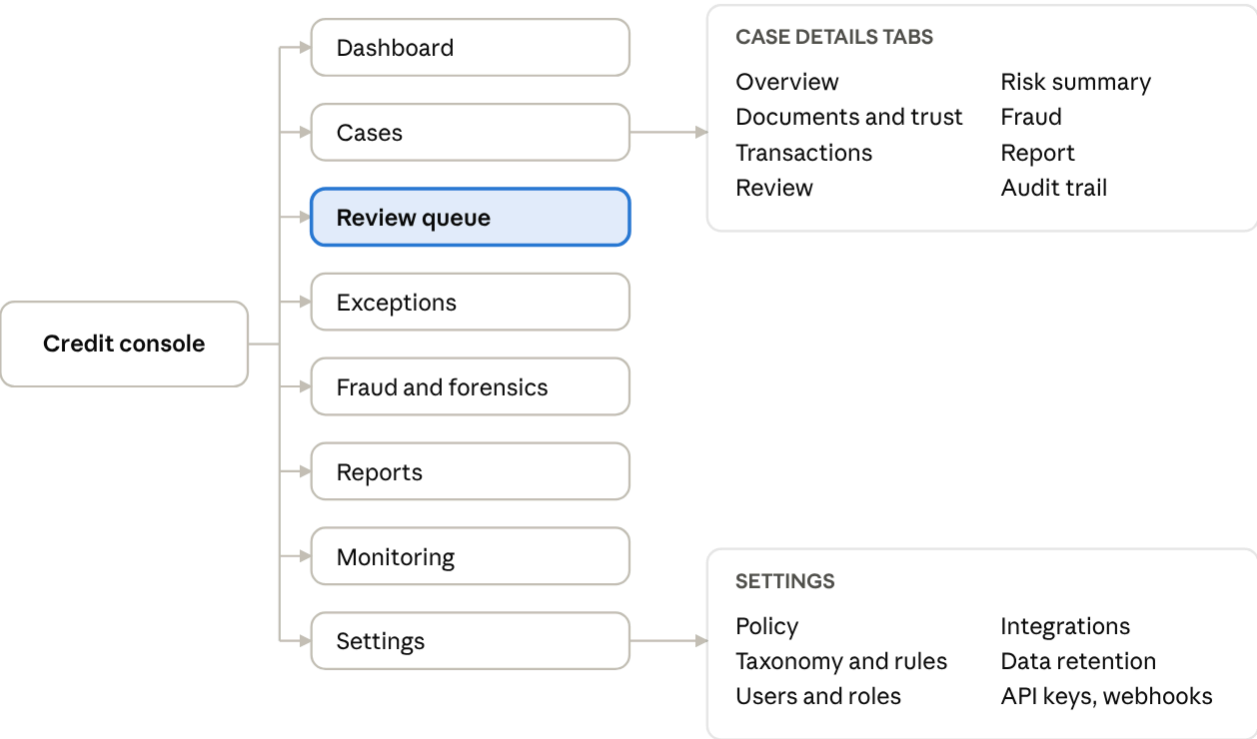
Group	KPI	Definition	Pilot target	12-month target
Risk	Early delinquency by band	30+ DPD at 6 months on book and 90+ DPD at 12 months, by score band	Data collection	Falls steadily from Poor to Excellent band
Business	Approval rate	APPROVE + CONDITIONS ÷ decided cases (watch, don't maximise)	Report	Stable within lender's appetite
Business	Cost per statement	(Compute + OCR + LLM tokens + storage) ÷ statements	Measure	≤ ₹5 [Approx.]
Business	Revenue and retention	Paying lenders; net revenue retention	1 pilot	8–10 lenders; NRR ≥ 110%
Platform	API uptime	Successful requests ÷ all, monthly	99.5%	99.9%
Platform	Webhook delivery	Delivered within 1 minute ÷ all	≥ 99%	≥ 99.9%

How to use them: quality KPIs gate releases (no parser release if accuracy drops on the labelled set); trust KPIs are reviewed weekly with the lender; risk KPIs are reviewed monthly and drive policy changes.

16. UX structure

Finding: organise the console around **the case**, because that is the unit of work officers, committees and auditors all share. Every case opens into the same eight tabs in the order an officer reads them: overview first, audit trail last. The **Review queue** is also a top-level area because officers work it across cases.

Eight top-level areas; every case opens into the same eight tabs



information architecture · 8 top-level areas, 8 case tabs, 6 settings

The prototype's seven screens (Home, Ingest, Transactions, Review queue, Risk summary, Activity log, Taxonomy) map onto these: Ingest becomes Documents and trust, Activity log becomes Audit trail, Taxonomy moves under Settings.

Screen by screen

Screen	Purpose	Key elements (top to bottom)	Primary actions	Main users
Dashboard	What needs me now; is the system healthy	Action bar (my cases, review items, SLA breaches); my queue; alerts; risk view adds KPI tiles and charts (section 7)	Open case; open queue	All

Screen	Purpose	Key elements (top to bottom)	Primary actions	Main users
Case list	Find and triage cases	Search; filters (stage, decision, product, bank, flag, DSA, date); table with SLA timer and decision chip	Open; assign; bulk export	Officer, ops
Case overview	Decide in two minutes	Header: applicant, product, amount, stage. Decision card (score, band, decision, top 3 reasons). Trust summary. Six component cards with ⓘ. Open review items count	Accept recommendation; propose override; send to committee	Officer
Documents and trust	Is the data usable and genuine?	Files with pages and source (upload / AA); validation checks grouped by layer with severity; page viewer	Supply password; reorder pages; request new statement	Ops, officer
Transactions	See every row and its label	Account and month filters; table with date, narration, amount, balance, category, confidence, method; monthly summary strip	Change category; create rule; export	Officer

Screen	Purpose	Key elements (top to bottom)	Primary actions	Main users
Review (case tab and global queue)	Decide only what matters	Items sorted by ₹ impact; each with narration, suggestion, why unsure, similar past decisions; score preview after a change	Accept; change; apply to counterparty; skip	Officer
Risk summary	Understand the score	Score and band; component cards with ⓘ (what it measures, how it is scored, this statement); hard rules that fired; policy version	Open calculation; compare with previous version	Officer, underwriter
Fraud (case tab and FCU area)	Investigate signals	Signals with evidence (page crops, metadata, money-flow chains); verification options; verdict form	Request AA pull; record verdict	FCU
Report	The committee and audit file	Preview of the PDF/Excel report; version selector	Download; send to LOS	Officer, committee
Audit trail	What happened, by whom, when	Timeline of events; version replay; export	Replay; export	Auditor, compliance
Exceptions	Unstick cases	STOP cases by reason; page image beside rows; re-run options	Re-run with LLM; fix header; return to DSA	Ops

Screen	Purpose	Key elements (top to bottom)	Primary actions	Main users
Reports (portfolio)	Is it working?	Accuracy, review load, decision mix, override analysis, band vs DPD	Filter; schedule; export	Risk head
Monitoring	After disbursal	Watchlist; alerts by severity; signal history per borrower	Acknowledge; assign to collections	Collections, risk
Settings	Configure the tenant	Policy (versions, simulate, publish); taxonomy and counterparty rules; users and roles; integrations; data retention; API keys and webhook logs	Edit with maker- checker	Admin, risk head

UX principles: one decision per screen; every number explains itself (i); show confidence and method beside machine-made labels; never hide what was overridden; responsive layout for DSAs on phones (the prototype already has mobile cards and a hamburger menu).

17. Competitive gap analysis

Finding: the poorly solved problems are not about *reading* statements. They are about **trusting, explaining and improving** the result: officers re-check what they cannot see into; risk teams cannot change policy themselves; review effort never falls; disagreements between PDF and AA data go unused. These are LedgerLens's openings.

Feature	Commonly available	Advanced (best in market)	Opportunity for LedgerLens	Priority
Explanation of the score	Variable list, long PDF report	Reason codes (Plaid LendScore)	Plain-language ⓘ on every number, using the borrower's own figures , the same text in UI, PDF and API	MVP
Review workflow	Manual check of the whole report	Human-in-the-loop extraction (Oculus)	Review only what moves the decision, sorted by ₹ impact , with score preview	MVP
Learning from officers	Vendor updates rules centrally	—	Officer fixes become tenant rules; accuracy per rule tracked; review effort shown falling	MVP
Lender-owned policy	Vendor score + lender's spreadsheet rules outside the tool	Decision engines sold separately	Versioned policy layer on transparent components, with simulation on past cases (no-code in Phase 2)	MVP
Decision replay for audit	Stored PDF report	—	Exact replay of any past result by version (parser, taxonomy, scorecard, policy)	MVP
Document forensics	Metadata and arithmetic checks (Perfios, FinBox)	Forensic + semantic + network checks, AI-generated detection (Inscribe)	Match the market, and add PDF-vs-AA cross-check : same account, same month, two sources	Phase 2
AA data	Fetch and analyse AA data	Multi-AA gateways with insights (Setu)	AA-first with upload fallback, one consistent result ; show which numbers came from which source	Phase 2

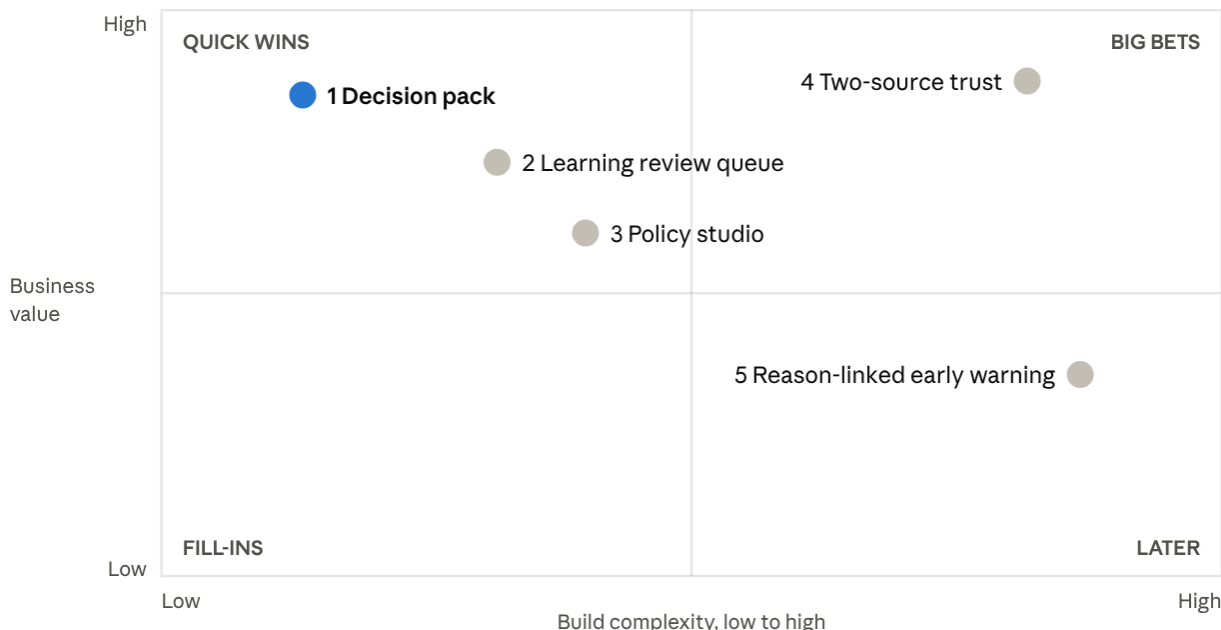
Feature	Commonly available	Advanced (best in market)	Opportunity for LedgerLens	Priority
MSME underwriting	Bank + GST + ITR combined report (ScoreMe, Perfios)	Packaged MSME scores	Turnover the officer can defend: GST-to-bank reconciliation with each gap explained	Phase 2
Multi-bank identity	Exact name match or none	Internal fuzzy matching	Visible fuzzy match with a confidence , and a clear REFER when it is really another person	Phase 2
Stress signals	Bounces and penalties	500+ parameters	A small set of named stress signals: loan-app stacking, minimum-due-only card payments, gambling or crypto, salary delay	Phase 2
Integration	REST API, PDF report	SDKs and LOS connectors	Webhooks, idempotency, sandbox with real-looking sample statements , case-level API	Phase 2
Post-disbursal	Re-pull on request	Continuous cash-flow monitoring (Plaid)	Early-warning alerts tied to the original decision's reasons ("the salary you approved on has stopped")	Phase 3
Officer assistant	—	AI fraud agents (Inscribe)	Ask the case: answers linked to the exact transactions; never decides	Phase 3
Score validation	Vendor-claimed accuracy	Model trained on repayment outcomes (Plaid LS1)	Band-vs-delinquency reports per lender , and champion/challenger before policy changes	Phase 3

Sources for market columns: section 1. Where a cell says "—" no product was found selling it in India in this research [Approx.].

18. Five differentiating bets

Finding: build bets 1–3 into the MVP and Phase 2. They reuse what the prototype already has (explanations, review queue, versioned results) and attack the officer's real pain. Bets 4 and 5 are bigger and depend on Account Aggregator data, so they follow.

Start with the decision pack: highest value for the least build



five bets placed by our judgement on value and build complexity [Assumption]

Positions are our judgement, not measured. Bet 1 is the only quick win; bets 4 and 5 need AA integration first.

Bet 1. The decision pack: explain and defend every decision

- **User problem:** officers re-check reports because they cannot see how numbers were made; committees and auditors get thin files; borrowers get no reason.
- **Workflow:** every case produces one pack: decision, top reasons with reason codes, each component's calculation with the borrower's own numbers, the trust summary, overrides, and the exact versions used. The same content appears in the UI (i), the PDF/Excel report and the API.
- **Business value:** cuts officer minutes per case (north-star metric); audit-ready by default; supports adverse-action reasons. Strong sales story: "you can defend every decision".
- **Complexity:** Low. Explanations already exist; add reason codes, the report template and version replay.

- **MVP approach:** reason codes for the 15 most common decision drivers; PDF report; replay endpoint. Measure officer handling time before and after in the pilot.

Bet 2. A review queue that learns

- **User problem:** review effort never falls. The same unclear employer or counterparty is checked again on every case.
- **Workflow:** queue shows only items that can move income, FOIR or a hard rule, sorted by ₹ impact. Each fix offers "apply to this counterparty"; the rule is tenant-scoped, versioned and reversible. A "what we learned" panel shows rules created, rows they settled, and review items saved per week.
- **Business value:** auto-decision rate rises month on month; visible proof of ROI for the lender; data for better default rules.
- **Complexity:** Medium. Impact sorting and counterparty rules exist; needs assignment, SLAs, rule quality tracking and safeguards against a bad rule spreading.
- **MVP approach:** impact sorting + counterparty rules + a weekly "review items saved" metric; rule approval by a senior user.

Bet 3. Policy studio: the lender owns its policy

- **User problem:** lenders' cut-offs live in spreadsheets or vendor tickets; changes take weeks; nobody knows the effect before go-live.
- **Workflow:** risk head edits knock-outs, cut-offs and band mappings per product; runs a simulation on the last 90 days ("112 decisions would change: 80 more CONDITIONS, 32 more DECLINE"); a checker approves; the version is published and stamped on every result.
- **Business value:** faster product launches; policy changes based on evidence; stickiness (policy lives in LedgerLens).
- **Complexity:** Medium. Needs a rule model, simulation over stored results, and maker-checker.
- **MVP approach:** policy as a versioned JSON config per tenant with a simulation report (MVP); a visual editor in Phase 2.

Bet 4. Two-source trust: PDF and AA cross-check

- **User problem:** forged PDFs pass simple checks; AI-generated statements are rising; FCU teams verify by phone and email.
- **Workflow:** when a borrower uploads a PDF, LedgerLens can request a lightweight AA consent for the same account and month. It compares balances and rows; any difference becomes a REFER with the exact rows. Document forensics (metadata, edits, fonts) runs on every PDF, and verified results train the fraud library.

- **Business value:** large cut in statement fraud and first-payment defaults; a trust score lenders can price on.
- **Complexity:** High. AA integration, consent UX, matching logic and forensics.
- **MVP approach (Phase 2):** metadata and incremental-edit checks first; PDF-vs-AA comparison for one pilot lender's AA flow. **[Reg/Legal]** consent purpose for verification.

Bet 5. Reason-linked early warning

- **User problem:** lenders find stress only when an EMI bounces; generic monitoring alerts are noisy.
- **Workflow:** at approval, the decision pack records *why* the loan was approved (e.g. salary ₹1.4 lakh from ACME, FOIR 28%). With monitoring consent, monthly AA data is checked against those same reasons: salary missing, FOIR now 55%, new loan-app EMIs. Alerts say which approval reason broke.
- **Business value:** earlier collections action; lower 30+ DPD; feedback to policy (which reasons break most often).
- **Complexity:** High. Needs AA periodic consent, monitoring jobs, alert routing to collections systems.
- **MVP approach (Phase 3):** three signals only (salary stop, new EMI, bounce) for one lender's book; compare 30+ DPD for alerted vs non-alerted loans.

19. Regulatory and legal register, open questions, sources

Finding: nothing here blocks the MVP, but **four items must be cleared with counsel before the pilot goes live:** LedgerLens's role under the Digital Lending Directions, data storage in India, LLM use on borrower data, and how rejection reasons are produced. Each is marked **[Reg/Legal]** in the sections above.

Register

#	Topic	What we know (source)	Product impact	Question for counsel	Before
R1	RBI Digital Lending Directions, 2025	Issued 8 May 2025; covers regulated entities and their Lending Service Providers (LSPs), including "services incidental to underwriting"; data stored in India, offshore-processed data returned within 24 hours; minimal data with LSPs (Lawrbit summary)	LedgeLens may count as an LSP or technology vendor to the lender; drives data storage, contracts and DLA reporting	Is LedgeLens an LSP? What contract clauses does the lender need?	Pilot
R2	Data localisation	Directions require storage in India (R1)	India cloud regions only; LLM calls must not leave India unless allowed	May masked data be processed by an LLM outside India if returned within 24 h?	Pilot
R3	DPDP Act 2023 and DPDP Rules 2025	Rules notified 13 Nov 2025, phased: consent-manager rules from Nov 2026; notice, consent, security, breach and retention rules from May 2027 (Deccan Herald)	Lender is the data fiduciary, LedgeLens a processor; retention limits, breach notice, deletion on request	Our processor obligations, retention period for raw statements, breach timelines	GA (May 2027 deadline)
R4	Account Aggregator framework	RBI-regulated AAs share data only with explicit consent for a stated purpose; GSTN is a data provider for GSTR-1/ 3B (Sahamati)	Separate consent purposes for underwriting, verification (bet 4) and monitoring (bet 5)	Can LedgeLens act as the lender's TSP, or must the lender be the FIU? Allowed purposes and retention for monitoring	Phase 2
R5	Fair Practices Code: rejection reasons	Lenders must convey in writing the main reasons for rejecting any loan application (RBI notification)	Reason codes and plain-language reasons (bet 1) support this; lender sends the letter	Are LedgeLens reason codes acceptable as the basis for rejection reasons?	Pilot

#	Topic	What we know (source)	Product impact	Question for counsel	Before
R6	Responsible AI (RBI FREE-AI framework)	RBI committee report of 13 Aug 2025: 7 principles ("sutras"), 26 recommendations, board-approved AI policies, AI-specific consumer protection and audits (KPMG summary)	LLM only for reading, never deciding; logs; per-tenant switch; model documentation for lenders' AI inventories	What documentation do lenders need from us for their AI policy and audits?	Pilot
R7	Automated credit decisions	Lender is responsible for the credit decision [Assumption]	Recommendation only; officer accepts; maker-checker for overrides	Is a fully automated APPROVE for small tickets permissible and advisable later?	Phase 3
R8	Credit bureau data	Bureau access is for members under credit-information law [Approx.]	Lender pulls the report and passes it in; LedgerLens does not pull bureaus	Can LedgerLens store bureau data received from the lender, and for how long?	Phase 2
R9	Fraud reporting and evidence	Lenders have their own fraud-reporting duties [Approx.]	Evidence pack and retention for confirmed forgeries	Retention period and format for fraud evidence	Phase 2
R10	Outsourcing and IT risk for regulated entities	Lenders apply outsourcing and IT controls to vendors [Approx.]	Security questionnaire, audits, right to audit, exit plan	Which certifications (ISO 27001, SOC 2) do target lenders require?	Pilot

Open questions for the business

- ☐ Confirm the first segment: NBFC personal loans of ₹50,000 to ₹5 lakh (section 1, assumption 1).
- ☐ Pricing model: per statement, per decided case, or platform fee plus usage?
- ☐ Will we offer a rules-only, no-LLM mode as a standard tier for cautious lenders?
- ☐ Which pilot lender can provide 300+ labelled historical statements with outcomes?
- ☐ Build or partner for Account Aggregator access (direct FIU integration vs a TSP gateway)?

Sources

Vendor pages and press could not be opened from this environment; claims come from search-result summaries of these pages and should be verified in vendor demos. As of 4 Oct 2026.

- [Perfios Bank Statement Analyser](#) · [Inc42: Perfios acquires Karza](#)
- [FinBox BankConnect](#) · [FinBox fraud page](#)
- [Outlook Startup: KreditBee and Digitap](#)
- [CB Insights: ScoreMe](#)
- [Setu Insights docs](#)
- [Unico Connect: BSA software in India](#)
- [Plaid Consumer Report](#) · [How Plaid built LendScore](#)
- [Ocrolus](#) · [Inscribe fake bank statement detector](#)
- [Business Standard: NBFCs lead AA consents in FY25](#)
- [Sahamati: ReBIT GSTN data schema](#)
- [Lawrbit: RBI Digital Lending Directions 2025](#)
- [Deccan Herald: DPDP Rules notified](#)
- [KPMG: RBI FREE-AI committee report](#)
- [RBI notification on Fair Practices Code](#)
- LedgerLens prototype facts: repository [bhawnamanglapm/LedgerLens1](#) (README sections 3b, 4, 5, 8).

Appendix: the brief this blueprint answers

The document above responds to the following brief, reproduced as written.

Act as a Senior Product Manager, Lending & Credit-Underwriting Product Architect, and UX strategist.

I am building LedgerLens, a Bank Statement Analysis (BSA) platform that helps lenders decide whether a loan can be given. A working prototype already exists: it ingests PDF / scanned / password-protected / CSV statements (multi-month, multi-account), classifies every transaction (rules → LLM → manual review), and produces a 1000-point credit risk score with a decision (APPROVE / APPROVE WITH CONDITIONS / REFER / DECLINE), a review queue, an activity log and an API with a background queue.

The goal is to understand what already exists in the market, identify gaps/opportunities, and design a differentiated, production-grade product – using the prototype as the starting point.

First, research and analyse leading bank-statement-analysis, cash-flow underwriting and account-aggregator players relevant to India (and global benchmarks where useful). Compare their capabilities across: statement ingestion, data extraction/OCR, Account Aggregator (AA) integration, transaction categorisation, income detection, obligation (EMI) detection, fraud/tamper detection, scoring/decisioning, reporting, APIs, LOS integrations, and operations.

Then create a complete product blueprint covering:

Product vision

What problem the platform solves (for lenders, credit officers, borrowers)

Target users/personas (credit officer, underwriter, risk head, ops/QC, DSA/sales, borrower, developer at the lender)

Key use cases (personal loan, business loan, credit card, BNPL, home loan, MSME, gig workers, NRI)

Value proposition

Differentiation opportunities

Current market capabilities

Identify commonly available features, including:

Statement upload (PDF, scanned, image, password-protected, Excel/CSV)

Account Aggregator / net-banking fetch

Multi-bank, multi-account, multi-month merging

Statement types: individual savings, salary, joint, current/business, credit card

OCR and table extraction

Data validation (page order, balance arithmetic, period coverage, duplicates)

Transaction categorisation

Salary / income detection (vs P2P, self-transfers, refunds, loan disbursements)

EMI / obligation detection and FOIR

Bounce, penalty and overdraft detection

Cash-flow and average-balance analysis

Fraud and tamper detection (edited PDFs, circular transactions, structuring, round-tripping)

Credit scoring and decision rules

Reports (PDF/Excel/JSON)

APIs/webhooks

LOS/LMS integrations

User and role management

Audit logs

Advanced/differentiating capabilities

Identify features that could make the platform significantly more valuable:

LLM-based extraction for unseen bank formats

Explainable scoring (why this score, in plain language)
 Document forensics (PDF metadata, font/edit detection)
 Business (MSME) cash-flow underwriting: turnover, GST vs bank credits, customer concentration, seasonality
 Gig / irregular income assessment
 Credit card behaviour (utilisation, minimum-due-only, cash withdrawals)
 Name/identity matching across banks
 Loan-app stacking and gambling/crypto signals
 Policy engine: configurable lender rules without code
 Champion/challenger scoring
 Early-warning / post-disbursal monitoring via AA consent
 AI underwriting assistant (ask questions about a statement in natural language)
 Bureau + statement cross-check (undeclared loans)
 Human-in-the-loop review with learning from corrections

End-to-end flow

Design the complete flow from:

Borrower → Lender/DSA → Upload or AA consent → Ingestion → Validation →
 Extraction → Enrichment → Classification → Human review → Scoring →
 Decision/Policy → LOS → Post-disbursal monitoring → Reporting.
 Explain each stage and the systems/components involved.

Product modules

Define the complete platform architecture, including:

Dashboard
 Case / application management
 Ingestion (upload + AA)
 Validation & integrity
 Transactions
 Review queue
 Risk scoring & decision
 Policy / rules engine
 Fraud & forensics
 Reports & analytics
 Monitoring (post-disbursal)
 Integrations (LOS, bureau, AA, GST)
 Notifications
 Users / roles & permissions
 Admin / ops console
 Developer / API portal

Dashboard

Design key metrics and widgets:
 Applications processed
 Auto-decision rate

Approval / refer / decline mix
 Average time to decision
 Review-queue backlog and SLA
 Extraction accuracy / OCR failures
 Fraud flags by type
 Bank/format coverage
 Income-band and FOIR distribution
 Alerts / action items

Operations console
 Design workflows for:
 Searching applications and transactions
 Fixing a failed or partial extraction
 Reviewing low-confidence transactions
 Overriding a category or decision (with reason)
 Investigating a fraud flag
 Handling password-protected or jumbled statements
 Escalating to a credit committee
 Viewing audit trails

Validation & integrity engine
 Design an automated system that checks statement trustworthiness:
 Balance arithmetic, opening/closing, page order, period coverage, duplicates across files, holder-name consistency, PDF tampering signals, exception handling, manual verification, validation reports.

Risk & decision engine
 Design a rule-based and AI-assisted engine covering:
 Income stability, debt service (FOIR), liquidity, banking behaviour, fraud indicators, expense management, hard overrides (no income, FOIR cap, tampering), segment-specific scorecards (salaried / self-employed / business / credit card), configurable thresholds, explainability, manual review and overrides.

API & technical architecture
 Propose an API-first architecture covering:
 Upload / AA-fetch APIs, job/status APIs, result APIs (versioned), review/decision APIs, webhooks, authentication, idempotency, job states, error handling, PII handling (masking, encryption, retention), LLM usage controls, monitoring, logging, scaling.

User journeys
 Create detailed journeys for:
 Credit officer analysing a salaried applicant
 Underwriting an MSME current account
 Applicant with several banks

Password-protected / scanned statement
 Tampered statement detected
 Low-confidence transactions in review
 Override and credit-committee escalation
 AA-based fetch with consent
 Post-disbursal early-warning alert

MVP

Prioritise features into MVP, Phase 2 and Phase 3 using MoSCoW, and explain why each belongs there. Clearly mark what the current LedgerLens prototype already covers.

PRD

Create a detailed PRD for the MVP with:

Problem statement, goals, non-goals, personas, user stories, functional requirements, non-functional requirements, acceptance criteria, success metrics, dependencies, risks, assumptions.

KPIs

Recommend product and business metrics such as:

Extraction accuracy, categorisation accuracy, auto-decision rate, time to decision, review effort per case, override rate, fraud catch rate, false-positive rate, bank-format coverage, API uptime, cost per statement, approval rate, early delinquency (30+/90+ DPD) of approved loans vs score band.

UX structure

Create a recommended information architecture and screen-by-screen structure: dashboard, case list, case details, ingestion, transactions, review queue, risk summary, fraud/forensics, reports, monitoring, settings.

Competitive gap analysis

Create a table: Feature | Commonly available | Advanced | Opportunity for LedgerLens | Priority

Do not simply copy existing BSA vendors. Identify genuine lender/credit-officer problems that are still poorly solved.

Finally, recommend 3-5 high-impact product ideas that could become LedgerLens's differentiating capabilities. For each, explain the user problem, workflow, expected business value, complexity, and MVP approach.

Assume the platform must be secure, scalable, API-first, enterprise-ready and suitable for India's regulatory environment. Clearly distinguish assumptions from confirmed requirements, and flag anything that needs regulatory/legal

validation (e.g. RBI Digital Lending Directions, Account Aggregator framework, DPDP Act data protection and consent, data localisation and retention, fair-practices code, explainability of automated credit decisions, use of LLMs on borrower data).